

Question 1

```
(avinash@kali)-[~]
$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
(avinash@kali)-[~]
$ touch file.txt
(avinash@kali)-[~]
$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
(avinash@kali)-[~]
$ rm f
rm: cannot remove 'f': No such file or directory
(avinash@kali)-[~]
$ rm file.txt
(avinash@kali)-[~]
$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
(avinash@kali)-[~]
$ mkdir foldername
(avinash@kali)-[~]
$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder foldername index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
(avinash@kali)-[~]
$ rm -r foldername
(avinash@kali)-[~]
$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
```

question 2

```
(avinash@kali)-[~]
$ echo "welcome to ethical hacking lab" >sample.txt

(avinash@kali)-[~]
$ echo "this is line 2" >> sample.txt

(avinash@kali)-[~]
$ echo "this is line 3" >> sample.txt

(avinash@kali)-[~]
$ echo "this is line 4" >> sample.txt

(avinash@kali)-[~]
$ less sample.txt

(avinash@kali)-[~]
$ more sample.txt
welcome to ethical hacking lab
this is line 2
this is line 3
this is line 4

(avinash@kali)-[~]
$ vi sample.txt

(avinash@kali)-[~]
$ cat sample.txt
welcome to ethical hacking lab
this is line 2
this is line 3
this is line 4this is adding using vi editor

(avinash@kali)-[~]
$ less sample.txt
```

question 3

```
(avinash@kali)-[~]
$ tree

192.168.1.10
Desktop
├── q01
│   └── Screenshot_2026-08-04_13_41_50.png
Documents
Downloads
ex1.txt
file2.txt
folder
index.html
Music
name
Pictures
├── Screenshot_2026-08-04_13_37_58.png (deleted)
├── Screenshot_2026-08-04_13_37_59.png (deleted)
├── Screenshot_2026-08-04_13_38_00.png (deleted)
├── Screenshot_2026-08-04_13_38_01.png (deleted)
└── Screenshot_2026-08-04_21_01_11.png
Projects
Public
sample.txt
Templates
testfolder
test.txt
Videos

14 directories, 12 files

(avinash@kali)-[~]
$ head sample.txt
welcome to ethical hacking lab
this is line 2
this is line 3
this is line 4this is adding using vi editor

(avinash@kali)-[~]
$ head -2 sample.txt
welcome to ethical hacking lab
this is line 2

(avinash@kali)-[~]
$ tail sample.txt
welcome to ethical hacking lab
this is line 2
this is line 3
this is line 4this is adding using vi editor

(avinash@kali)-[~]
$ tail -2 sample.txt
this is line 3
this is line 4this is adding using vi editor
```

question 4

```
(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos

(avinash@kali)~$ ls -l
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
Desktop
Documents
Downloads
ex1.txt
file2.txt
folder
index.html
Music
name
Pictures
Projects
Public
sample.txt
Templates
testfolder
test.txt
Videos

(avinash@kali)~$ ls -a
. .bashrc.original Documents file2.txt .java Pictures .ssh .vboxclient-clipboard-tty7-control.pid .vboxclient-seamless-tty7-control.pid .viminfo .zsh_history
.cache Downloads folder .local .profile .sudo_as_admin_successful .vboxclient-clipboard-tty7-service.pid .vboxclient-seamless-tty7-service.pid .zshrc
192.168.1.10 .config ex1.txt .gnupg .mozilla Projects Templates .vboxclient-draganddrop-tty7-control.pid .vboxclient-vmsvga-session-tty7-control.pid .xsession-errors
.bash_logout Desktop .face .ICAuthority Music Public testfolder .vboxclient-draganddrop-tty7-service.pid .vboxclient-vmsvga-session-tty7-service.pid .xsession-errors.old
.bashrc .dmrc .face.icon index.html name sample.txt test.txt .vboxclient-hostversion-tty7-control.pid Videos .zprofile

(avinash@kali)~$ cat sample.txt
cat: sample.txt: No such file or directory

(avinash@kali)~$ cat > file.txt
hello
ethical hacking lab

(avinash@kali)~$ cat file1.txt
cat: file1.txt: No such file or directory

(avinash@kali)~$ mkdir testfolder
mkdir: cannot create directory 'testfolder': File exists

(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
```

question 5

```
(avinash@kali)~$ uptime
21:52:20 up 5:25, 1 user, load average: 0.04, 0.08, 0.11

(avinash@kali)~$ date
Tuesday 04 August 2026 09:52:25 PM IST

(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos

(avinash@kali)~$ mv file1.txt file2.txt
mv: cannot stat 'file1.txt': No such file or directory

(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos

(avinash@kali)~$ cat file2.txt
hello
ethical hacking lab
```

question 6

```
(avinash@kali)~$ touch test.txt

(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos

(avinash@kali)~$ pwd
/home/avinash

(avinash@kali)~$ wget https://example.com
--2026-08-04 21:54:59-- https://example.com/
Resolving example.com (example.com)... 2606:4700:83b5:72db:f2c8:494:ef6b:ff98, 104.20.23.154, 172.66.147.243
Connecting to example.com (example.com)[2606:4700:83b5:72db:f2c8:494:ef6b:ff98]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html.1'

index.html.1 [ => ] 559 --.-KB/s in 0s

2026-08-04 21:55:00 (3.26 MB/s) - 'index.html.1' saved [559]

(avinash@kali)~$ ls
192.168.1.10 Desktop Documents Downloads ex1.txt file2.txt file.txt folder index.html index.html.1 Music name Pictures Projects Public sample.txt Templates testfolder test.txt Videos
```

Question 7

```
(avinash@kali:~)$ cd ~
(avinash@kali:~)$ pwd
/home/avinash
(avinash@kali:~)$ cp file2.txt copy.txt
(avinash@kali:~)$ ls
192.168.1.10  copy.txt  Desktop  Documents  Downloads  ex1.txt  file2.txt  file.txt  folder  index.html  index.html.1  Music  name  Pictures  Projects  Public  sample.txt  Templates  testfolder  test.txt  Videos
(avinash@kali:~)$ cat copy.txt
hello
ethical hacking lab
(avinash@kali:~)$ ls
192.168.1.10  copy.txt  Desktop  Documents  Downloads  ex1.txt  file2.txt  file.txt  folder  index.html  index.html.1  Music  name  Pictures  Projects  Public  sample.txt  Templates  testfolder  test.txt  Videos
(avinash@kali:~)$ ls -l
192.168.1.10
copy.txt
Desktop
Documents
Downloads
ex1.txt
file2.txt
file.txt
folder
index.html
index.html.1
Music
name
Pictures
Projects
Public
sample.txt
Templates
testfolder
test.txt
Videos
(avinash@kali:~)$ ls -la
.          .cache  Downloads  folder  .local  Projects  testfolder  .vboxclient-hostversion-tty7-control.pid  .viminfo  .zshrc
..         .config  ex1.txt    .gnupg  .mozilla  Public    test.txt   .vboxclient-seamless-tty7-control.pid  .xauthority
192.168.1.10  copy.txt  .face     .ICAuthority  Music    sample.txt  .vboxclient-clipboard-tty7-control.pid  .vboxclient-seamless-tty7-service.pid  .xsession-errors
.bash_logout  .dirc   file2.txt  index.html  name     .ssh       .vboxclient-clipboard-tty7-service.pid  .vboxclient-vmsvga-session-tty7-control.pid  .xsession-errors.old
.bashrc       .dmrc   file2.txt  index.html.1  Pictures  .sudo_as_admin_successful  .vboxclient-draganddrop-tty7-control.pid  .vboxclient-vmsvga-session-tty7-service.pid  .zprofile
.bashrc.original  Documents  file.txt   .java      .profile  Templates  .vboxclient-draganddrop-tty7-service.pid  Videos  .zsh_history
```

question 8

```
(avinash@kali:~)$ id addr
id: 'addr': no such user
(avinash@kali:~)$ nmap -sV 192.168.1.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 21:59 +0530
Failed to resolve "192.168.1.10".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.71 seconds
(avinash@kali:~)$ nmap -A 192.168.1.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 21:59 +0530
Nmap scan report for 192.168.1.10
Host is up (0.050s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: power-device/firewall|WAP/router|general purpose|specialized
Running (JUST GUESSING): APC embedded (94%), Cisco ASA 9.X (94%), Cisco embedded (94%), Synology embedded (94%), Microsoft Windows 2000/2003 (93%), ETH Zurich Bluebottle (87%), Tibbo embedded (86%)
OS CPE: cpe:/o:cisco:asa:9.2 cpe:/h:synology:rt1900ac cpe:/o:microsoft:windows_2000::sp4 cpe:/o:microsoft:windows_server_2003::sp2 cpe:/o:ethzurich:bluebottle
Aggressive OS guesses: APC Network Management Card 3 (94%), Cisco Adaptive Security Appliance (ASA 9.2) (94%), Cisco Aironet 3800-series WAP (94%), Synology RT1900ac router (94%), Microsoft Windows 2000 SP4 (93%), Microsoft Windows Ser
ver 2003 SP2 (93%), Bluebottle OS (87%), Tibbo DS10xx serial controller, V3.34 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 6 hops

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 6.93 ms 10.154.71.108
2 ... 5
6 42.93 ms 192.168.1.10

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.47 seconds
(avinash@kali:~)$ sudo nmap -o 192.168.1.10
[sudo] password for avinash:
/usr/lib/nmap/nmap: unrecognized option '-o'
See the output of nmap -h for a summary of options.
```


Question 9

```
(avinash@kali)-[~]
$ nmap -T2 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 22:02 +0530
Stats: 0:00:42 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 10.50% done; ETC: 22:08 (0:06:07 remaining)
Stats: 0:00:44 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 11.10% done; ETC: 22:08 (0:06:00 remaining)
Stats: 0:02:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.10% done; ETC: 22:08 (0:04:10 remaining)
Stats: 0:06:38 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.20% done; ETC: 22:08 (0:00:03 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000063s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 401.20 seconds

(avinash@kali)-[~]
$ nmap -T3 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 22:09 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

Question 10

```
(avinash@kali)-[~]
$ free
              total        used        free      shared  buff/cache   available
Mem:           4011596      1129784      1730420         25812       1434000       2881012
Swap:              0         2719740              0

(avinash@kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi       1.1Gi       1.6Gi       25Mi       1.4Gi       2.7Gi
Swap:           2.6Gi           0B         2.6Gi

(avinash@kali)-[~]
$ cat > names.txt
pravi
anil
kumar
pala

(avinash@kali)-[~]
$ sort names.txt
anil
pala
pravi
kumar

(avinash@kali)-[~]
$ history
1 jhfk1gherhglghiuufuyegfsuygdtxgougw8efpphv8efwzeuEWGF
2 LCFLGHAOHGARGOPWRFOGweflbg;WENF9EWHFR
3 nmap -sv 192.168.1.1
4 198.185.159.144
5 rm file.txt
6 rm -r foldername
7 ls
8 ls
9 touch file.txt
10 ls
11 f
12 rm file.txt
13 rm -r foldername
14 mkdir foldername
15 ls
16 rm
17 rm -r foldername
18 ls
19 touch file.txt
20 ls
21 rm file.txt
22 mkdir foldername
23 ls
24 rm -r foldername
25 ls
26 echo "welcome to ethical hacking lab"
27 sample.txt
```